Allison Newman

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Education

Cornell University (GPA 3.93), *Bachelor of Science, Mechanical Engineering* **Aug. 2024 - Exp. May 2027**

Relevant courses: Statics, Intro to Aeronautics, Mechanics of Materials, Fluid Mechanics, Thermodynamics, Dynamics, Intro to Lasers and Photonics

Experiences

Lockheed Martin RMS–Sikorsky, *Assembly & Flight Operations, Manufacturing Eng. Intern* **May - Jul. 2026**

- Simulated and modeled gas-purging process using inert gas for AFSD with STAR-CCM+ and Ansys to demonstrate inefficiencies in current setup and suggest improvements for future systems. Familiarized myself with steady, transient and turbulent models; implemented passive scalar models.
- Sliced, printed, and packaged 26 to-scale parts of the modernized Black Hawk nose for a trade-show in under 2 weeks utilizing 3 F900 printers.
- Designed fixtures to assist in stamping part-numbers onto welding substrates, removing laser powderbed fusion parts, and depositing welding rods.
- Conceptualized propulsion system for hybrid VTOL aircraft capable of providing 18HP and hovering for 20 minutes; awarded 2nd place. Performed trade study on motor & engine. Chose rotor and ESC to accompany motors; created test & safety guide for ground and flight tests.

Cornell University Unmanned Air Systems (CUAir), *Structures and Payloads Engineer* **Nov. 2024 - Present**

- Dedicating 9+ hours weekly to help design, manufacture, and test a carbon-fiber-composite search-and-rescue autonomous aircraft as part of the CUASC competition. Performed carbon-fiber layups of fuselage and bulkhead; collecting data with Integration and Testing from wing-loading.
- Placed second in flight and design at CUASC UAV 2025 international competition. Team routinely places first in innovation at SUAS UAV comp.
- Prototyped a novel tilt-rotor system for a 35lb eVTOL aircraft, optimizing for DFMA and integration with Autopilot and Electrical.

Dewey Smart, *Academic and Test Prep Tutor, College Counselor* **May. 2025 - Present**

- Contributing 6+ hours weekly to instructing and preparing students for College Applications, in addition to academic tutoring in Calculus I and II.
- Clarifying key math and english concepts assessed by the SAT, improving score by 200+ points; researching and assigning practice worksheets.

Academic Excellence Workshop, *Facilitator for MATH2940 - Linear Algebra* **Aug. 2025 - Present**

- Curating 2-hour weekly workshops with co-facilitator to reinforce key Linear Algebra principles for undergraduate students; drafting, proofreading, and testing worksheets and games to engage students while highlighting key concepts, totaling 4+ hours of prep weekly.
- Working closely with Linear Algebra TAs and Office of Inclusive Excellence to establish a challenging and reflective workshop to improve students' understanding of difficult concepts by attending biweekly training to reinforce active learning principles built off of group collaboration.

Clubs and Volunteering

Society of Women Engineers, *Communications Codirector, Alumni & Faculty Committee Member* **Sep. 2024 - Present**

- Contacted over 20 professors at Cornell University to secure a diverse group of mentors for the annual student-faculty dinner with 30+ attendees.
- Curated a themed social media page dedicated to spreading awareness about opportunities for women in engineering.

Projects

Tilt Rotor Mechanism for 35lb eVTOL Aircraft, *CUAir Project*

- Designed a tilt rotor mechanism to provide 45lbs of thrust, capable of flying at 25m/s. Enables transitioning between flight modes at 180 deg./ sec.
- Performed statics and dynamics hand calcs on boom, clevis, and gear geometry to ensure a high factor of safety for bearing and bending stress.
- Modeled in SolidWorks linkage vs. gear design to determine optimal rotation mechanics; performed FEA on booms and brackets for deflection.

Object-Collecting Robot, *Mechatronics Project*

- Ranked 3rd out of 60 teams for collecting the most blocks against another robot opponent; collected up to 15 blocks in <1 min.
- Designed in SolidWorks using COTS + bespoke parts to be fabricated out of acrylic; assembled hands-on with laser-cutter, dremel, and superglue.
- Programmed Arduino using C++ and digital registers. Utilized 3 servos and 2 QTIs + color sensor for board localization/ border detection.

Honors and Awards

Asian American Government Executives Network, *Annual Scholarship* **2025**

Recognized with a $2000 scholarship for academic and leadership background that aligns with AAGEN's mission to promote Asian Americans in local, state, and federal government positions. The award is bestowed on individuals with "the potential and motivation to bring significant change to leadership in the public sector".

California State Board of Education, *State Seal of Civic Engagement* **2024**

Awarded for community garden project for demonstrating civic skills and action with a testimony of civic mindedness; one of 6 people in high school of 2000 to receive this state-recognized award.

Skills and Certifications

Technical: SolidWorks bent sheet metal, CATIA, 3DX, STAR-CCM+ CFD, Ansys FEA, MATLAB, Java, Arduino, GrabCAD Print, Insight, AFSD, FDM 3D Printing, GD&T, DFMA, Rapid Prototyping, Shaper CNC, Composite Layups & Testing, Mill, Lathe

Professional: Microsoft Suite, Google Workspace, Confluence, Solumina, Project Management, OSHA Safety, ESH, FOD Awareness